Nischal Bhandari

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Full-Stack Software Engineer | System Design · Data Modeling · Python · TypeScript · React

SUMMARY

Full-stack engineer who designs and architects systems end-to-end — data models, multi-system integrations, security boundaries, and production automation — with correctness verified, not assumed. 1.5 years at Four Hands building internal platforms in Python, TypeScript, React, and PostgreSQL.

EXPERIENCE

Four Hands — Austin, TX Jul 2025 – Present
Software Engineer (Platform & Internal Tooling)

- **FH Connect Hub — system architecture:** designed and built the internal EDI platform that unified five systems (support mailbox, Jira, Confluence, Mentis customer data, and an AI assistant) into one app. Architected the data flow — Logic App ingest → Azure SQL → queued background analysis workers → human review → Jira ticket → Confluence note — with Entra ID auth and scoped Key Vault secrets.
- **fh-d365-mcp — security architecture:** designed a read-only MCP gateway for Dynamics 365: a policy layer that rejects any non-approved host or write intent *before* a token is attached, TLS verification always on, zero stored secrets — with blast radius and the migration to a service-principal identity documented.
- **AI test-generation pipeline — system design:** designed a pipeline with deterministic gates (tag check → content-sufficiency check → human approval) before any code generation, plus failure diagnosis that reads real execution logs to separate a broken test from a broken application.
- **Data modeling & reliability:** modeled a normalized domain schema (client → account → position → order → execution) with migrations and JWT ownership checks; own CI/CD and observability — 500+ E2E tests, telemetry, and error tracking across async workers.

ZT Systems (acquired by AMD) — Georgetown, TX Jun 2024 – Sep 2024
System Test Engineer / QA Analyst

- Automated RAID/BIOS server validation in Python + Bash; root-cause failure analysis on 75+ hardware configuration issues.

PROJECTS

StockTradingApp — *Python, FastAPI, SQLAlchemy, PostgreSQL*
Domain-first data model: normalized relational schema with table-per-subtype orders, Alembic migrations, and ownership-enforced APIs.

QA Studio / RecorderApp — *TypeScript, React, Electron, Playwright*
Workspace-based plugin architecture: recorder → locator registry → codegen pipeline, with deterministic output for reviewable diffs.

Account Insights — *Salesforce LWC, Apex, GraphQL*
Snapshot worker built from chained, checkpointed jobs with run tokens and per-page data-accuracy rules; scores reconciled exact against production.

d365fo-client — *Python, MCP*
Read-only-by-default MCP server with a request policy gate in front of every call.

SKILLS

Design & Architecture: system design, data modeling, API design, security (least-privilege), deterministic verification
Languages: Python, TypeScript, JavaScript, SQL, Apex
Frontend / Backend: React, Angular, Next.js, LWC · FastAPI, Node.js, Express, GraphQL
Data: PostgreSQL, SQLAlchemy, Azure SQL, Alembic
Cloud, CI/CD & Testing: Docker, Linux, Azure, GitHub Actions, Azure DevOps, Playwright, BrowserStack

EDUCATION

B.S. Computer Science *(Minor in Mathematics)* — Texas State University Graduated 2025